

# Networks and Causal Inference Short Course: Part 3

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## Using Data from the Transmission Reduction Intervention Project in Athens, Greece

### Download the data and load it into R Studio

This is a simulated toy dataset made for this course. It is based on data collected from the Transmission Reduction Intervention Project in Athens Greece. Public data here: <https://www.icpsr.umich.edu/web/NAHDAP/studies/39059/versions/V2/staff>

Read more: <https://pubmed.ncbi.nlm.nih.gov/27917890/>

Load packages and function script...

```
# -----  
# 1. Read data and functions into R  
# -----  
  
# Set the working directory to the current file's folder  
# install.packages("rstudioapi")  
library(rstudioapi)  
setwd(dirname(rstudioapi::getActiveDocumentContext()$path))  
getwd() # verify the change
```

```
## [1] "/Users/buchanan/Library/CloudStorage/OneDrive-UniversityofRhodeIsland/R01 MSM HIV Networks/Train
```

```
library(lme4)  
library(plyr)  
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.2
```

```
library(igraph)  
library(numDeriv)  
library(gtools)  
library(doParallel)  
library(foreach)  
library(knitr)  
library(kableExtra)  
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

```
library(forcats)  
rm(list = ls())
```

```
no_cores <- detectCores() - 1  
registerDoParallel(cores=no_cores)
```

```
#Can source GitHub page
#source("https://github.com/uri-ncipher/Nearest-Neighbor-estimators/blob/main/functions.R?raw=TRUE")
source("../code/IPW_functions.R")
source("../code/gcomp_dr_v2.R") #removed warnings for bootstrap
```

## Data Preparation

Read in the synthetic nodes and edges data sets for creating the simulated network.

```
# -----
# 2. Data Preparation
# -----

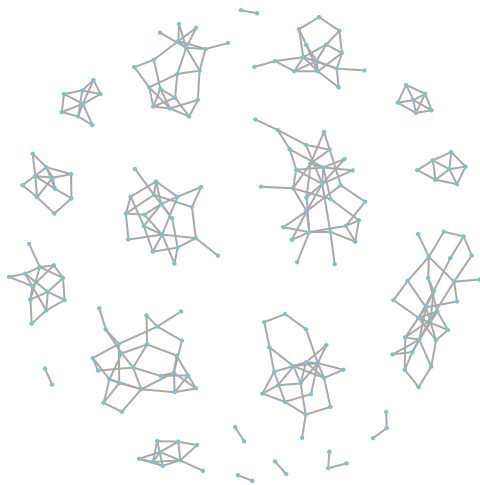
nodes <- read.csv("../TRIP_network_simulation/TRIPsim_nodes.csv")
edges <- read.csv("../TRIP_network_simulation/TRIPsim_edges.csv")

net0=graph_from_data_frame(d=edges, vertices = nodes, directed=F) # create network based on the nodes
```

## Network visualization

```
# -----
# 3. Network Visualization
# -----

# plot the network generated above
plot(net0, vertex.size=1, vertex.label=NA, vertex.color="cadetblue3", vertex.frame.color="cadetblue3")
```



## Prepare the data for modeling

```
# -----
# 3. Data Preparation for Modeling
# -----

# Save the number of individuals (nodes) and number of components
n=length(V(net0)) # number of individuals
m=components(net0)$no # number of components

# generate a dataset with three columns, the first column is participant id (node id),
```

```

# the second column is number of nearest neighbors for each participant (is also node degree),
# the third column indicate which component the participant is in.

df=data.frame(id=1:n, na=unlist(lapply(1:n, num_neighbors, net=net0)),
              component=components(net0)$membership)

# assign treatment, outcome, and baseline covariates to data
df$treatment=nodes$treatment
df$outcome=nodes$outcome
df$age=nodes$age
df$age10=nodes$age/10 #age per 10 years
df$agebin=ifelse(nodes$age>35,1,0) #age >35 years old
df$posneg=nodes$posneg
df$date=nodes$date
df$share=nodes$share
df$education=nodes$education
df$employment=nodes$employment
df$na_a= unlist(lapply(1:n, trt_neighbors, net=net0))
df$notna_a=df$na-df$na_a

# five covariates, excluding age
base_covariate=c("posneg", "date", "share", "education", "employment")

df$avg_posneg = nodes$avg_posneg
df$avg_date = nodes$avg_date
df$avg_share = nodes$avg_share
df$avg_education = nodes$avg_education
df$avg_employment = nodes$avg_employment

avg_covariate=c("avg_posneg", "avg_date", "avg_share", "avg_education","avg_employment")

```

## Modeling

In this section, we will apply both IPW1 and IPW2 estimators for the average potential outcomes and corresponding causal effects. An individual's exposure/treatment and allocation strategy  $\alpha$  for their contacts jointly determine the average potential outcome. Allocation strategy represents the probability that individuals in nearest neighbors set receiving the exposure/treatment in the counterfactual scenario, so it is a numeric value between 0 and 1. For each estimator, the R programs will output the **point estimation and the estimated variance of average potential outcomes**  $\hat{Y}(1, \alpha)$ ,  $\hat{Y}(0, \alpha)$ ,  $\hat{Y}(\alpha)$  under three different allocation strategies  $\alpha$  (i.e., 0.25, 0.50, and 0.75), respectively. Also, the R program will output the **point estimation and estimated variance of four causal effects: Direct (DE), Indirect (IE), Total (TE), and Overall effect (OE)**.

Equations for calculating the four causal effects are:

- $\widehat{DE} = \hat{Y}(1, \alpha) - \hat{Y}(0, \alpha)$
- $\widehat{IE} = \hat{Y}(0, \alpha_0) - \hat{Y}(0, \alpha_1)$
- $\widehat{TE} = \hat{Y}(1, \alpha_0) - \hat{Y}(0, \alpha_1)$
- $\widehat{OE} = \hat{Y}(\alpha_0) - \hat{Y}(\alpha_1)$ ,

where  $\alpha_0$ ,  $\alpha_1$  both represents allocation strategies, and  $\alpha_0 \neq \alpha_1$  (Note: The associated paper compared the estimated average potential outcomes with  $\alpha_1$  to  $\alpha_0$  for the indirect, total, and overall effects. The code here compared the estimated average potential outcomes with  $\alpha_0$  to  $\alpha_1$  for the indirect, total, and overall effects). Users can set any values between 0 and 1 for  $\alpha_0$  and  $\alpha_1$ . If given  $\alpha$  is a vector, then the causal contrasts will

produce pairwise comparisons in the sequential order of the values in  $\alpha$ . For example, if  $\alpha = c(0.75, 0.25, 0.50)$ , the contrasts for spillover effects will compare:

- $\alpha_0 = 0.75$  versus  $\alpha_1 = 0.25$ ;
- $\alpha_0 = 0.75$  versus  $\alpha_1 = 0.50$ ;
- $\alpha_0 = 0.25$  versus  $\alpha_1 = 0.50$ .

## IPW1

Using IPW1 to estimate the average potential outcomes and causal effects under allocation strategies  $\alpha$ .

```
# -----
# 4a. IPW1 Estimator
# -----

alpha=c(0.5, 0.4, 0.3, 0.2) # set allocation strategies

base_covariate=c("posneg", "date", "share", "education", "employment")
avg_covariate=c("avg_posneg", "avg_date", "avg_share", "avg_education", "avg_employment")

IPW1 <- IPW_1_model(df, base_covariate, alpha)
IPW1
```

```
## [[1]]
##           a=1           a=0           margin alpha           type
## 1 0.238639343 0.345578481 0.292108912 0.5 point estimate
## 2 0.330162950 0.394691853 0.368880292 0.4 point estimate
## 3 0.442566828 0.436954921 0.438638493 0.3 point estimate
## 4 0.562156764 0.466673389 0.485770064 0.2 point estimate
## 5 0.008475355 0.005133750 0.005093127 0.5      variance
## 6 0.017711096 0.005925071 0.007865991 0.4      variance
## 7 0.032419413 0.006980926 0.010366261 0.3      variance
## 8 0.047891472 0.008009177 0.011289713 0.2      variance
##
## [[2]]
##           estimation alpha0 alpha1           type
## 1 -0.1069391376      0.5      0.5 Direct
## 2 -0.0645289032      0.4      0.4 Direct
## 3  0.0056119063      0.3      0.3 Direct
## 4  0.0954833748      0.2      0.2 Direct
## 5  0.0068457040      0.5      0.5 Var DE
## 6  0.0115562094      0.4      0.4 Var DE
## 7  0.0202200560      0.3      0.3 Var DE
## 8  0.0293495214      0.2      0.2 Var DE
## 9 -0.0491133720      0.5      0.4 Indirect
## 10 -0.0913764406      0.5      0.3 Indirect
## 11 -0.1210949086      0.5      0.2 Indirect
## 12 -0.0422630687      0.4      0.3 Indirect
## 13 -0.0719815367      0.4      0.2 Indirect
## 14 -0.0297184680      0.3      0.2 Indirect
## 15  0.0003328171      0.5      0.4 Var IE
## 16  0.0012632802      0.5      0.3 Var IE
## 17  0.0025340800      0.5      0.2 Var IE
## 18  0.0003081928      0.4      0.3 Var IE
## 19  0.0010826964      0.4      0.2 Var IE
```

```
## 20  0.0002482060    0.3    0.2  Var IE
## 21 -0.1560525096    0.5    0.4   Total
## 22 -0.1983155782    0.5    0.3   Total
## 23 -0.2280340462    0.5    0.2   Total
## 24 -0.1067919718    0.4    0.3   Total
## 25 -0.1365104398    0.4    0.2   Total
## 26 -0.0241065617    0.3    0.2   Total
## 27  0.0055407193    0.5    0.4  Var TE
## 28  0.0048145106    0.5    0.3  Var TE
## 29  0.0047336534    0.5    0.2  Var TE
## 30  0.0101693297    0.4    0.3  Var TE
## 31  0.0096168144    0.4    0.2  Var TE
## 32  0.0191242809    0.3    0.2  Var TE
## 33 -0.0767713795    0.5    0.4 Overall
## 34 -0.1465295813    0.5    0.3 Overall
## 35 -0.1936611524    0.5    0.2 Overall
## 36 -0.0697582018    0.4    0.3 Overall
## 37 -0.1168897729    0.4    0.2 Overall
## 38 -0.0471315711    0.3    0.2 Overall
## 39  0.0004243409    0.5    0.4  Var OE
## 40  0.0013439552    0.5    0.3  Var OE
## 41  0.0021361214    0.5    0.2  Var OE
## 42  0.0002840771    0.4    0.3  Var OE
## 43  0.0008691923    0.4    0.2  Var OE
## 44  0.0002365923    0.3    0.2  Var OE
```

In the output above,

- The **section** `[[1]]` displays the point estimates (type = “point estimate” in the output) and estimated variances (type = “variance” in the output) for the average potential outcomes under three different allocation strategies (i.e.,  $\alpha = 0.25, 0.50, 0.75$ ) and each individual exposure ( $a = 0, a = 1$ , and margin). “Margin” is the average potential outcomes for a particular allocation strategy, regardless of individual exposure status.
- The **section** `[[2]]` displays the point estimates (type = “Direct”, “Indirect”, “Total”, “Overall” in the output) and estimated variances (type = “Var DE”, “Var IE”, “Var TE”, “Var OE” in the output) of four causal effects: direct, indirect, total and overall effects, corresponding to particular allocation strategies ( $\alpha_0$  and  $\alpha_1$ ). The estimated values are shown in “estimation” column in the output above.

## IPW2

Using IPW2 to estimate the average potential outcomes and causal effects under allocation strategies  $\alpha$ .

```
# -----
# 4b. IPW2 Estimator
# -----

base_covariate=c("posneg", "date", "share", "education", "employment")
avg_covariate=c("avg_posneg", "avg_date", "avg_share", "avg_education", "avg_employment")

alpha=c(0.5, 0.4, 0.3, 0.2) # set allocation strategies

formula_1=paste("cbind(na_a, notna_a)", "~", "treatment", "+",
               paste(base_covariate, collapse = "+"), "+",
               paste(avg_covariate, collapse = "+"))
formula_2=paste("treatment", "~", paste(base_covariate, collapse = "+"))
M1=glm(formula_1, family = binomial(link = "logit"), data=df)
```

```
M2=glm(formula_2, family = binomial(link = "logit"), data=df)
```

```
IPW2 <- IPW_2_model(df, M1, M2, alpha)
```

```
IPW2
```

```
## [[1]]
```

| ##   |             | a=1        | a=0        | margin | alpha          | type |
|------|-------------|------------|------------|--------|----------------|------|
| ## 1 | 0.225756485 | 0.57259834 | 0.39917741 | 0.5    | point estimate |      |
| ## 2 | 0.298721919 | 0.57364088 | 0.46367330 | 0.4    | point estimate |      |
| ## 3 | 0.394876037 | 0.55878647 | 0.50961334 | 0.3    | point estimate |      |
| ## 4 | 0.512422291 | 0.53221910 | 0.52825974 | 0.2    | point estimate |      |
| ## 5 | 0.006352282 | 0.08669419 | 0.02879003 | 0.5    | variance       |      |
| ## 6 | 0.010834579 | 0.05452511 | 0.02772639 | 0.4    | variance       |      |
| ## 7 | 0.020404295 | 0.02892717 | 0.02256607 | 0.3    | variance       |      |
| ## 8 | 0.038624594 | 0.01502655 | 0.01690969 | 0.2    | variance       |      |

```
##
```

```
## [[2]]
```

| ##    | estimation    | alpha0 | alpha1 | type     |
|-------|---------------|--------|--------|----------|
| ## 1  | -0.3468418569 | 0.5    | 0.5    | Direct   |
| ## 2  | -0.2749189618 | 0.4    | 0.4    | Direct   |
| ## 3  | -0.1639104357 | 0.3    | 0.3    | Direct   |
| ## 4  | -0.0197968061 | 0.2    | 0.2    | Direct   |
| ## 5  | 0.0709328137  | 0.5    | 0.5    | Var DE   |
| ## 6  | 0.0388437767  | 0.4    | 0.4    | Var DE   |
| ## 7  | 0.0181154162  | 0.3    | 0.3    | Var DE   |
| ## 8  | 0.0177279203  | 0.2    | 0.2    | Var DE   |
| ## 9  | -0.0010425395 | 0.5    | 0.4    | Indirect |
| ## 10 | 0.0138118687  | 0.5    | 0.3    | Indirect |
| ## 11 | 0.0403792438  | 0.5    | 0.2    | Indirect |
| ## 12 | 0.0148544082  | 0.4    | 0.3    | Indirect |
| ## 13 | 0.0414217834  | 0.4    | 0.2    | Indirect |
| ## 14 | 0.0265673752  | 0.3    | 0.2    | Indirect |
| ## 15 | 0.0047106238  | 0.5    | 0.4    | Var IE   |
| ## 16 | 0.0209995034  | 0.5    | 0.3    | Var IE   |
| ## 17 | 0.0459575391  | 0.5    | 0.2    | Var IE   |
| ## 18 | 0.0058272656  | 0.4    | 0.3    | Var IE   |
| ## 19 | 0.0212940414  | 0.4    | 0.2    | Var IE   |
| ## 20 | 0.0048549053  | 0.3    | 0.2    | Var IE   |
| ## 21 | -0.3478843964 | 0.5    | 0.4    | Total    |
| ## 22 | -0.3330299882 | 0.5    | 0.3    | Total    |
| ## 23 | -0.3064626131 | 0.5    | 0.2    | Total    |
| ## 24 | -0.2600645536 | 0.4    | 0.3    | Total    |
| ## 25 | -0.2334971784 | 0.4    | 0.2    | Total    |
| ## 26 | -0.1373430606 | 0.3    | 0.2    | Total    |
| ## 27 | 0.0407602304  | 0.5    | 0.4    | Var TE   |
| ## 28 | 0.0178068702  | 0.5    | 0.3    | Var TE   |
| ## 29 | 0.0066802414  | 0.5    | 0.2    | Var TE   |
| ## 30 | 0.0165682125  | 0.4    | 0.3    | Var TE   |
| ## 31 | 0.0062452709  | 0.4    | 0.2    | Var TE   |
| ## 32 | 0.0088316681  | 0.3    | 0.2    | Var TE   |
| ## 33 | -0.0644958833 | 0.5    | 0.4    | Overall  |
| ## 34 | -0.1104359291 | 0.5    | 0.3    | Overall  |
| ## 35 | -0.1290823234 | 0.5    | 0.2    | Overall  |
| ## 36 | -0.0459400458 | 0.4    | 0.3    | Overall  |

```
## 37 -0.0645864401    0.4    0.2 Overall
## 38 -0.0186463943    0.3    0.2 Overall
## 39  0.0005131657    0.5    0.4 Var OE
## 40  0.0032122374    0.5    0.3 Var OE
## 41  0.0097388734    0.5    0.2 Var OE
## 42  0.0013019590    0.4    0.3 Var OE
## 43  0.0064352776    0.4    0.2 Var OE
## 44  0.0020154208    0.3    0.2 Var OE
```

In the output above,

- The **section** `[[1]]` displays the point estimates (type = “point estimate” in the output) and estimated variances (type = “variance” in the output) for the average potential outcomes under three different allocation strategies (i.e.,  $\alpha = 0.75, 0.50, 0.25$ ) and each individual exposure ( $a = 0, a = 1$ , and margin). “Margin” is the average potential outcomes for a particular allocation strategy, regardless of individual exposure status.
- The **section** `[[2]]` displays the point estimates (type = “Direct”, “Indirect”, “Total”, “Overall” in the output) and estimated variances (type = “Var DE”, “Var IE”, “Var TE”, “Var OE” in the output) of four causal effects: direct, indirect, total and overall effects, corresponding to particular allocation strategies ( $\alpha_0$  and  $\alpha_1$ ). The estimated values are shown in “estimation” column in the output above.

Now, we provide example code for an outcome-model based approach, g-computation. We use assumptions similar to IPW2 regarding the exposure and covariates for confounding from an individual’s contacts in the network. We obtain 95% confidence intervals using a bootstrap on each individual and the summary information for their contacts’ exposures and covariates.

```
# -----
# 4c. G-computation Estimator (similar assumptions to IPW2)
# -----

#Obtain G-computation estimates with bootstrap CIs
#Uses similar assumptions for interference as IPW2
#Bootstrap unit is each individual and the information for their contacts

alpha <- c(0.5, 0.4, 0.3, 0.2)

res_gcomp <- gcomp_with_bootstrap(
  data = df,
  alpha = alpha,
  z_covariates = base_covariate,
  zn_covariates = avg_covariate,
  B = 1000,
  seed = 2026
)

res_gcomp$results
```

```
##      estimation alpha0 alpha1      type      lower      upper
## 1 -0.2671042126    0.5    0.5 Direct -0.48990800  0.17435337
## 2 -0.2048266478    0.4    0.4 Direct -0.40569963  0.15317102
## 3 -0.1387097545    0.3    0.3 Direct -0.32371304  0.13187768
## 4 -0.0688200772    0.2    0.2 Direct -0.24681674  0.15040116
## 5  0.0257653426    0.5    0.5 Var DE          NA          NA
## 6  0.0190595871    0.4    0.4 Var DE          NA          NA
## 7  0.0135269355    0.3    0.3 Var DE          NA          NA
## 8  0.0099261178    0.2    0.2 Var DE          NA          NA
```

|       |               |     |     |          |             |            |
|-------|---------------|-----|-----|----------|-------------|------------|
| ## 9  | 0.0202681181  | 0.5 | 0.4 | Indirect | -0.01892687 | 0.04460728 |
| ## 10 | 0.0407533916  | 0.5 | 0.3 | Indirect | -0.03806904 | 0.09361120 |
| ## 11 | 0.0614054455  | 0.5 | 0.2 | Indirect | -0.05739987 | 0.14550949 |
| ## 12 | 0.0204852735  | 0.4 | 0.3 | Indirect | -0.01914363 | 0.04860563 |
| ## 13 | 0.0411373274  | 0.4 | 0.2 | Indirect | -0.03847837 | 0.10041133 |
| ## 14 | 0.0206520539  | 0.3 | 0.2 | Indirect | -0.01933474 | 0.05183156 |
| ## 15 | 0.0002714276  | 0.5 | 0.4 | Var IE   | NA          | NA         |
| ## 16 | 0.0011456427  | 0.5 | 0.3 | Var IE   | NA          | NA         |
| ## 17 | 0.0027101895  | 0.5 | 0.2 | Var IE   | NA          | NA         |
| ## 18 | 0.0003020533  | 0.4 | 0.3 | Var IE   | NA          | NA         |
| ## 19 | 0.0012677649  | 0.4 | 0.2 | Var IE   | NA          | NA         |
| ## 20 | 0.0003324272  | 0.3 | 0.2 | Var IE   | NA          | NA         |
| ## 21 | -0.2468360945 | 0.5 | 0.4 | Total    | -0.45373988 | 0.17233550 |
| ## 22 | -0.2263508210 | 0.5 | 0.3 | Total    | -0.41886543 | 0.17024901 |
| ## 23 | -0.2056987671 | 0.5 | 0.2 | Total    | -0.38780374 | 0.17681363 |
| ## 24 | -0.1843413744 | 0.4 | 0.3 | Total    | -0.37240518 | 0.15560619 |
| ## 25 | -0.1636893205 | 0.4 | 0.2 | Total    | -0.34585162 | 0.15523127 |
| ## 26 | -0.1180577006 | 0.3 | 0.2 | Total    | -0.29954481 | 0.13509485 |
| ## 27 | 0.0233584479  | 0.5 | 0.4 | Var TE   | NA          | NA         |
| ## 28 | 0.0213879253  | 0.5 | 0.3 | Var TE   | NA          | NA         |
| ## 29 | 0.0199526903  | 0.5 | 0.2 | Var TE   | NA          | NA         |
| ## 30 | 0.0171761005  | 0.4 | 0.3 | Var TE   | NA          | NA         |
| ## 31 | 0.0158311097  | 0.4 | 0.2 | Var TE   | NA          | NA         |
| ## 32 | 0.0122853754  | 0.3 | 0.2 | Var TE   | NA          | NA         |
| ## 33 | -0.0313533290 | 0.5 | 0.4 | Overall  | -0.05321265 | 0.03334116 |
| ## 34 | -0.0511857883 | 0.5 | 0.3 | Overall  | -0.09808794 | 0.06668230 |
| ## 35 | -0.0583826454 | 0.5 | 0.2 | Overall  | -0.13097797 | 0.09522079 |
| ## 36 | -0.0198324593 | 0.4 | 0.3 | Overall  | -0.04509961 | 0.03195720 |
| ## 37 | -0.0270293163 | 0.4 | 0.2 | Overall  | -0.07980348 | 0.06225665 |
| ## 38 | -0.0071968570 | 0.3 | 0.2 | Overall  | -0.03676776 | 0.03415579 |
| ## 39 | 0.0004678267  | 0.5 | 0.4 | Var OE   | NA          | NA         |
| ## 40 | 0.0016478527  | 0.5 | 0.3 | Var OE   | NA          | NA         |
| ## 41 | 0.0032447664  | 0.5 | 0.2 | Var OE   | NA          | NA         |
| ## 42 | 0.0003727172  | 0.4 | 0.3 | Var OE   | NA          | NA         |
| ## 43 | 0.0013370938  | 0.4 | 0.2 | Var OE   | NA          | NA         |
| ## 44 | 0.0003179714  | 0.3 | 0.2 | Var OE   | NA          | NA         |

Now, we provide example code for an doubly-robust approach, augmented IPW. We use assumptions similar to IPW2 regarding the covariates for confounding from an individual's contacts in the network. We obtain 95% confidence intervals using a bootstrap on each individual and the summary information for their contacts' exposures and covariates.

```
# -----
# 4d. Augmented IPW Estimator (doubly robust (DR1))
# -----

#Obtain DR1 Estimates with Bootstrap CIs
#Uses similar assumptions for interference as IPW2
#Bootstrap unit is each individual and the information for their contacts

alpha=c(0.5, 0.4, 0.3, 0.2) # set allocation strategies

res_dr <- dr_with_bootstrap(
  data = df,
  alpha = alpha,
```

```

z_covariates = base_covariate,
zn_covariates = avg_covariate,
B = 1000,
seed = 2026
)
res_dr$results

```

| ##    | estimation    | alpha0 | alpha1 | type     | lower       | upper      |
|-------|---------------|--------|--------|----------|-------------|------------|
| ## 1  | -0.4528773483 | 0.5    | 0.5    | Direct   | -0.93833640 | 0.06757263 |
| ## 2  | -0.3548705097 | 0.4    | 0.4    | Direct   | -0.76379337 | 0.09345008 |
| ## 3  | -0.2417824557 | 0.3    | 0.3    | Direct   | -0.55369909 | 0.12890305 |
| ## 4  | -0.1153061958 | 0.2    | 0.2    | Direct   | -0.39119692 | 0.22426833 |
| ## 5  | 0.0611163683  | 0.5    | 0.5    | Var DE   | NA          | NA         |
| ## 6  | 0.0435031839  | 0.4    | 0.4    | Var DE   | NA          | NA         |
| ## 7  | 0.0328302948  | 0.3    | 0.3    | Var DE   | NA          | NA         |
| ## 8  | 0.0336135585  | 0.2    | 0.2    | Var DE   | NA          | NA         |
| ## 9  | 0.0462157119  | 0.5    | 0.4    | Indirect | -0.01309415 | 0.14991841 |
| ## 10 | 0.0939853833  | 0.5    | 0.3    | Indirect | -0.02883467 | 0.31682229 |
| ## 11 | 0.1398168289  | 0.5    | 0.2    | Indirect | -0.04679717 | 0.47328714 |
| ## 12 | 0.0477696714  | 0.4    | 0.3    | Indirect | -0.01606602 | 0.16697646 |
| ## 13 | 0.0936011170  | 0.4    | 0.2    | Indirect | -0.03283277 | 0.32364491 |
| ## 14 | 0.0458314456  | 0.3    | 0.2    | Indirect | -0.01683130 | 0.15670717 |
| ## 15 | 0.0017969958  | 0.5    | 0.4    | Var IE   | NA          | NA         |
| ## 16 | 0.0080955317  | 0.5    | 0.3    | Var IE   | NA          | NA         |
| ## 17 | 0.0182584119  | 0.5    | 0.2    | Var IE   | NA          | NA         |
| ## 18 | 0.0022733297  | 0.4    | 0.3    | Var IE   | NA          | NA         |
| ## 19 | 0.0086463231  | 0.4    | 0.2    | Var IE   | NA          | NA         |
| ## 20 | 0.0020638977  | 0.3    | 0.2    | Var IE   | NA          | NA         |
| ## 21 | -0.4066616364 | 0.5    | 0.4    | Total    | -0.82444705 | 0.06962704 |
| ## 22 | -0.3588919650 | 0.5    | 0.3    | Total    | -0.68210320 | 0.09351320 |
| ## 23 | -0.3130605194 | 0.5    | 0.2    | Total    | -0.57697032 | 0.11843591 |
| ## 24 | -0.3071008383 | 0.4    | 0.3    | Total    | -0.62092429 | 0.09873412 |
| ## 25 | -0.2612693927 | 0.4    | 0.2    | Total    | -0.51242752 | 0.13360680 |
| ## 26 | -0.1959510101 | 0.3    | 0.2    | Total    | -0.44825897 | 0.16298237 |
| ## 27 | 0.0473346039  | 0.5    | 0.4    | Var TE   | NA          | NA         |
| ## 28 | 0.0360347566  | 0.5    | 0.3    | Var TE   | NA          | NA         |
| ## 29 | 0.0293431941  | 0.5    | 0.2    | Var TE   | NA          | NA         |
| ## 30 | 0.0320782921  | 0.4    | 0.3    | Var TE   | NA          | NA         |
| ## 31 | 0.0252684900  | 0.4    | 0.2    | Var TE   | NA          | NA         |
| ## 32 | 0.0258624495  | 0.3    | 0.2    | Var TE   | NA          | NA         |
| ## 33 | -0.0382747583 | 0.5    | 0.4    | Overall  | -0.08184047 | 0.03797556 |
| ## 34 | -0.0599185542 | 0.5    | 0.3    | Overall  | -0.14444857 | 0.08231932 |
| ## 35 | -0.0635606061 | 0.5    | 0.2    | Overall  | -0.18489710 | 0.14792888 |
| ## 36 | -0.0216437958 | 0.4    | 0.3    | Overall  | -0.06467477 | 0.05037374 |
| ## 37 | -0.0252858478 | 0.4    | 0.2    | Overall  | -0.11320194 | 0.11860818 |
| ## 38 | -0.0036420519 | 0.3    | 0.2    | Overall  | -0.04964731 | 0.07353754 |
| ## 39 | 0.0008497344  | 0.5    | 0.4    | Var OE   | NA          | NA         |
| ## 40 | 0.0032990100  | 0.5    | 0.3    | Var OE   | NA          | NA         |
| ## 41 | 0.0073155951  | 0.5    | 0.2    | Var OE   | NA          | NA         |
| ## 42 | 0.0008981373  | 0.4    | 0.3    | Var OE   | NA          | NA         |
| ## 43 | 0.0036822376  | 0.4    | 0.2    | Var OE   | NA          | NA         |
| ## 44 | 0.0010126945  | 0.3    | 0.2    | Var OE   | NA          | NA         |

Lastly, we perform assessment of the propensity score in the presence of interference.

```

# -----
# 5. Assessing the fit of the generalized propensity scores
# -----

### Check covariate balance for generalized propensity score: Fit a model on each covariate, then include

#### check propensity score balance for IPW2
mat_1 = as.matrix(cbind(1, df[, row.names(coef(summary(M1)))[-1]])) # neighbors' treatment
mat_2 = as.matrix(cbind(1, df[, row.names(coef(summary(M2)))[-1]])) # Individual treatment

group_ps <- plogis(mat_1 %*% coef(M1))
ind_ps <- plogis(mat_2 %*% coef(M2))
df$group_ps_ipw2 <- group_ps
df$ind_ps_ipw2 <- ind_ps #Ke, should this be 1-ps for those not exposed?

### Step 1: check individual-level propensity score balance
# Note: Compare two models w/ and w/o treatment:
#      (1) ind_covariate ~ ind_treatment + ind_ps_ipw2;
#      (2) ind_covariate ~ ind_ps_ipw2

ipw2_balance <- data.frame()
for (i in 1:length(base_covariate)) {
  f1 <- paste0(base_covariate[i], " ~ ind_ps_ipw2")
  f2 <- paste0(base_covariate[i], " ~ treatment + ind_ps_ipw2")

  if (base_covariate[i] == "age10"){
    model_reduced <- glm(f1, data = df, family = gaussian)
    model_full <- glm(f2, data = df, family = gaussian("identity"))
  } else {
    model_reduced <- glm(f1, data = df, family = binomial("logit"))
    model_full <- glm(f2, data = df, family = binomial("logit"))
  }
  res_temp <- lmtest::lrtest(model_reduced, model_full)

  ipw2_balance <- rbind(ipw2_balance,
                        c(f1, f2, round(res_temp$Chisq[2], 4), round(res_temp$`Pr(>Chisq)`[2], 4))
  )
}

### Step 2: check group-level propensity score balance
# Note: Compare two models w/ and w/o treatment:
#      (1) group_covariate ~ group_treatment + group_ps;
#      (2) group_covariate ~ group_ps.

for (i in 1:length(avg_covariate)) {
  f1 <- paste0(avg_covariate[i], " ~ group_ps_ipw2")
  f2 <- paste0(avg_covariate[i], " ~ cbind(na_a, notna_a) + group_ps_ipw2")

  model_reduced <- glm(f1, data = df)
  model_full <- glm(f2, data = df)

  res_temp <- lmtest::lrtest(model_reduced, model_full)
}

```

```

    ipw2_balance <- rbind(ipw2_balance,
                          c(f1, f2, round(res_temp$Chisq[2], 4), round(res_temp$`Pr(>Chisq)`[2], 4))
    )
  }
  colnames(ipw2_balance) <- c("Reduced model", "Full model", "Chisq", "Pr(>Chisq)")
  ipw2_balance

##               Reduced model
## 1          posneg ~ ind_ps_ipw2
## 2              date ~ ind_ps_ipw2
## 3              share ~ ind_ps_ipw2
## 4      education ~ ind_ps_ipw2
## 5      employment ~ ind_ps_ipw2
## 6      avg_posneg ~ group_ps_ipw2
## 7      avg_date ~ group_ps_ipw2
## 8      avg_share ~ group_ps_ipw2
## 9      avg_education ~ group_ps_ipw2
## 10 avg_employment ~ group_ps_ipw2
##
##               Full model   Chisq   Pr(>Chisq)
## 1          posneg ~ treatment + ind_ps_ipw2 3e-04    0.9851
## 2              date ~ treatment + ind_ps_ipw2 2e-04    0.9899
## 3              share ~ treatment + ind_ps_ipw2 0.0103    0.9192
## 4      education ~ treatment + ind_ps_ipw2 0.0011    0.9735
## 5      employment ~ treatment + ind_ps_ipw2 0.0062    0.9371
## 6      avg_posneg ~ cbind(na_a, notna_a) + group_ps_ipw2 0.5768    0.7495
## 7      avg_date ~ cbind(na_a, notna_a) + group_ps_ipw2 1.3707    0.5039
## 8      avg_share ~ cbind(na_a, notna_a) + group_ps_ipw2 5.6176    0.0603
## 9      avg_education ~ cbind(na_a, notna_a) + group_ps_ipw2 0.4509    0.7982
## 10 avg_employment ~ cbind(na_a, notna_a) + group_ps_ipw2 2.3596    0.3073

#Optional: Format Latex Table for output of each estimation approach
#Create forest plot
library(dplyr)
library(forcats)
library(ggplot2)
library(knitr)
library(kableExtra)

z <- qnorm(0.975)

#####
# 1. Result object helper
#####

#conflicts_prefer(dplyr::filter) # filter will now always use dplyr, even if MASS is loaded
#conflicts_prefer(dplyr::count)

make_result_object <- function(x) {
  if (is.list(x) && !is.data.frame(x)) {
    df_idx <- which(sapply(x, function(y) {
      is.data.frame(y) &&
      all(c("estimation", "alpha0", "alpha1", "type") %in% names(y))
    }))
  }
}

```

```

    if (length(df_idx) == 0) {
      stop("No data frame with columns estimation, alpha0, alpha1, and type was found.")
    }

    x <- x[[df_idx[1]]]
  }

  if (!all(c("estimation", "alpha0", "alpha1", "type") %in% names(x))) {
    stop("Input must contain columns: estimation, alpha0, alpha1, type.")
  }

  est_rows <- x %>%
    filter(!grepl("^Var", type)) %>%
    mutate(
      effect_key = case_when(
        type == "Direct" ~ "DE",
        type == "Indirect" ~ "IE",
        type == "Total" ~ "TE",
        type == "Overall" ~ "OE",
        TRUE ~ type
      )
    )

  var_rows <- x %>%
    filter(grepl("^Var", type)) %>%
    mutate(
      effect_key = gsub("^Var\\s+", "", type),
      variance = estimation
    ) %>%
    select(alpha0, alpha1, effect_key, variance)

  results <- est_rows %>%
    left_join(
      var_rows,
      by = c("alpha0", "alpha1", "effect_key")
    ) %>%
    mutate(
      se = sqrt(variance),
      lower = estimation - z * se,
      upper = estimation + z * se
    ) %>%
    select(estimation, alpha0, alpha1, type, lower, upper)

  list(results = results)
}

#####
# 2. Standardize alpha display by estimator/effect
#####

prep_effect_results <- function(results, estimator_name) {
  out <- results %>%
    filter(!grepl("^Var", type)) %>%

```

```

mutate(
  estimator = estimator_name,
  effect = case_when(
    type == "Direct" ~ "Direct",
    type == "Indirect" ~ "Indirect",
    type == "Total" ~ "Total",
    type == "Overall" ~ "Overall",
    TRUE ~ type
  )
)

# G-comp and DR store coverage columns reversed relative to display.
if (estimator_name %in% c("G-comp", "DR")) {
  out <- out %>%
    mutate(
      raw_alpha1 = alpha0,
      raw_alpha0 = alpha1
    )
} else {
  out <- out %>%
    mutate(
      raw_alpha1 = alpha1,
      raw_alpha0 = alpha0
    )
}

out %>%
  mutate(
    display_alpha1 = pmax(raw_alpha1, raw_alpha0),
    display_alpha0 = pmin(raw_alpha1, raw_alpha0),
    contrast = paste0("(", display_alpha1, ", ", display_alpha0, ")")
  ) %>%
  filter(
    case_when(
      effect == "Direct" ~ display_alpha1 == display_alpha0,
      effect %in% c("Indirect", "Total") ~ display_alpha1 > display_alpha0,
      effect == "Overall" ~ TRUE,
      TRUE ~ TRUE
    )
  )
}

#####
# 3. LaTeX table helper
#####

make_latex_effect_table <- function(results,
                                     estimator_name,
                                     caption = "Estimated causal effects",
                                     digits = 3) {
  table_df <- prep_effect_results(results, estimator_name) %>%
    mutate(
      effect = factor(effect, levels = c("Direct", "Indirect", "Total", "Overall")),

```

```

estimate = round(estimation, digits),
ci = ifelse(
  is.na(lower) | is.na(upper),
  "",
  paste0(
    "(", round(lower, digits), ", ",
    round(upper, digits), ")"
  )
)
) %>%
arrange(effect, desc(display_alpha1), desc(display_alpha0)) %>%
select(
  Effect = effect,
  `$(\\alpha_1, \\alpha_0)$` = contrast,
  `Point estimate` = estimate,
  `95\\% CI` = ci
)

direct_n <- sum(table_df$Effect == "Direct")
indirect_n <- sum(table_df$Effect == "Indirect")
total_n <- sum(table_df$Effect == "Total")
overall_n <- sum(table_df$Effect == "Overall")

tab <- table_df %>%
  kable(
    format = "latex",
    booktabs = TRUE,
    escape = FALSE,
    caption = caption,
    align = c("l", "c", "c", "c")
  ) %>%
  kable_styling(latex_options = c("hold_position"))

start <- 1

if (direct_n > 0) {
  tab <- tab %>% pack_rows("Direct effects", start, start + direct_n - 1, italic = TRUE)
  start <- start + direct_n
}

if (indirect_n > 0) {
  tab <- tab %>% pack_rows("Indirect effects", start, start + indirect_n - 1, italic = TRUE)
  start <- start + indirect_n
}

if (total_n > 0) {
  tab <- tab %>% pack_rows("Total effects", start, start + total_n - 1, italic = TRUE)
  start <- start + total_n
}

if (overall_n > 0) {
  tab <- tab %>% pack_rows("Overall effects", start, start + overall_n - 1, italic = TRUE)
}

```

```

    tab
  }

#####
# 4. Create result objects
#####

res_ipw1 <- make_result_object(IPW1)
res_ipw2 <- make_result_object(IPW2)

res_gcomp <- if (exists("res_gcomp") && "results" %in% names(res_gcomp)) {
  res_gcomp
} else {
  make_result_object(gcomp_table)
}

res_dr <- if (exists("res_dr") && "results" %in% names(res_dr)) {
  res_dr
} else {
  make_result_object(dr_table)
}

#####
# 5. Create LaTeX tables
#####

ipw1_tab <- make_latex_effect_table(
  res_ipw1$results,
  estimator_name = "IPW1",
  caption = "IPW1 estimates of causal effects"
)

ipw2_tab <- make_latex_effect_table(
  res_ipw2$results,
  estimator_name = "IPW2",
  caption = "IPW2 estimates of causal effects"
)

gcomp_tab <- make_latex_effect_table(
  res_gcomp$results,
  estimator_name = "G-comp",
  caption = "G-computation estimates of causal effects"
)

dr_tab <- make_latex_effect_table(
  res_dr$results,
  estimator_name = "DR",
  caption = "Doubly robust estimates of causal effects"
)

cat(ipw1_tab, file = "ipw1_effects_table.tex")
cat(ipw2_tab, file = "ipw2_effects_table.tex")
cat(gcomp_tab, file = "gcomp_effects_table.tex")

```

```

cat(dr_tab, file = "dr_effects_table.tex")

#####
# 6. Forest plot data
#####

estimator_levels <- c("IPW1", "IPW2", "G-comp", "DR")

df <- bind_rows(
  prep_effect_results(res_ipw1$results, "IPW1"),
  prep_effect_results(res_ipw2$results, "IPW2"),
  prep_effect_results(res_gcomp$results, "G-comp"),
  prep_effect_results(res_dr$results, "DR")
) %>%
  mutate(
    estimator = factor(estimator, levels = estimator_levels),
    effect = factor(effect, levels = c("Direct", "Indirect", "Total", "Overall"))
  )

# Optional check: verify all estimator/effect/contrast combinations are present.
df %>%
  count(estimator, effect, contrast) %>%
  arrange(estimator, effect, contrast)

```

```

##   estimator   effect   contrast n
## 1      IPW1   Direct (0.2, 0.2) 1
## 2      IPW1   Direct (0.3, 0.3) 1
## 3      IPW1   Direct (0.4, 0.4) 1
## 4      IPW1   Direct (0.5, 0.5) 1
## 5      IPW1 Indirect (0.3, 0.2) 1
## 6      IPW1 Indirect (0.4, 0.2) 1
## 7      IPW1 Indirect (0.4, 0.3) 1
## 8      IPW1 Indirect (0.5, 0.2) 1
## 9      IPW1 Indirect (0.5, 0.3) 1
## 10     IPW1 Indirect (0.5, 0.4) 1
## 11     IPW1    Total (0.3, 0.2) 1
## 12     IPW1    Total (0.4, 0.2) 1
## 13     IPW1    Total (0.4, 0.3) 1
## 14     IPW1    Total (0.5, 0.2) 1
## 15     IPW1    Total (0.5, 0.3) 1
## 16     IPW1    Total (0.5, 0.4) 1
## 17     IPW1 Overall (0.3, 0.2) 1
## 18     IPW1 Overall (0.4, 0.2) 1
## 19     IPW1 Overall (0.4, 0.3) 1
## 20     IPW1 Overall (0.5, 0.2) 1
## 21     IPW1 Overall (0.5, 0.3) 1
## 22     IPW1 Overall (0.5, 0.4) 1
## 23     IPW2   Direct (0.2, 0.2) 1
## 24     IPW2   Direct (0.3, 0.3) 1
## 25     IPW2   Direct (0.4, 0.4) 1
## 26     IPW2   Direct (0.5, 0.5) 1
## 27     IPW2 Indirect (0.3, 0.2) 1
## 28     IPW2 Indirect (0.4, 0.2) 1
## 29     IPW2 Indirect (0.4, 0.3) 1

```

```

## 30      IPW2 Indirect (0.5, 0.2) 1
## 31      IPW2 Indirect (0.5, 0.3) 1
## 32      IPW2 Indirect (0.5, 0.4) 1
## 33      IPW2      Total (0.3, 0.2) 1
## 34      IPW2      Total (0.4, 0.2) 1
## 35      IPW2      Total (0.4, 0.3) 1
## 36      IPW2      Total (0.5, 0.2) 1
## 37      IPW2      Total (0.5, 0.3) 1
## 38      IPW2      Total (0.5, 0.4) 1
## 39      IPW2 Overall (0.3, 0.2) 1
## 40      IPW2 Overall (0.4, 0.2) 1
## 41      IPW2 Overall (0.4, 0.3) 1
## 42      IPW2 Overall (0.5, 0.2) 1
## 43      IPW2 Overall (0.5, 0.3) 1
## 44      IPW2 Overall (0.5, 0.4) 1
## 45      G-comp Direct (0.2, 0.2) 1
## 46      G-comp Direct (0.3, 0.3) 1
## 47      G-comp Direct (0.4, 0.4) 1
## 48      G-comp Direct (0.5, 0.5) 1
## 49      G-comp Indirect (0.3, 0.2) 1
## 50      G-comp Indirect (0.4, 0.2) 1
## 51      G-comp Indirect (0.4, 0.3) 1
## 52      G-comp Indirect (0.5, 0.2) 1
## 53      G-comp Indirect (0.5, 0.3) 1
## 54      G-comp Indirect (0.5, 0.4) 1
## 55      G-comp      Total (0.3, 0.2) 1
## 56      G-comp      Total (0.4, 0.2) 1
## 57      G-comp      Total (0.4, 0.3) 1
## 58      G-comp      Total (0.5, 0.2) 1
## 59      G-comp      Total (0.5, 0.3) 1
## 60      G-comp      Total (0.5, 0.4) 1
## 61      G-comp Overall (0.3, 0.2) 1
## 62      G-comp Overall (0.4, 0.2) 1
## 63      G-comp Overall (0.4, 0.3) 1
## 64      G-comp Overall (0.5, 0.2) 1
## 65      G-comp Overall (0.5, 0.3) 1
## 66      G-comp Overall (0.5, 0.4) 1
## 67      DR      Direct (0.2, 0.2) 1
## 68      DR      Direct (0.3, 0.3) 1
## 69      DR      Direct (0.4, 0.4) 1
## 70      DR      Direct (0.5, 0.5) 1
## 71      DR Indirect (0.3, 0.2) 1
## 72      DR Indirect (0.4, 0.2) 1
## 73      DR Indirect (0.4, 0.3) 1
## 74      DR Indirect (0.5, 0.2) 1
## 75      DR Indirect (0.5, 0.3) 1
## 76      DR Indirect (0.5, 0.4) 1
## 77      DR      Total (0.3, 0.2) 1
## 78      DR      Total (0.4, 0.2) 1
## 79      DR      Total (0.4, 0.3) 1
## 80      DR      Total (0.5, 0.2) 1
## 81      DR      Total (0.5, 0.3) 1
## 82      DR      Total (0.5, 0.4) 1
## 83      DR Overall (0.3, 0.2) 1

```

```

## 84      DR Overall (0.4, 0.2) 1
## 85      DR Overall (0.4, 0.3) 1
## 86      DR Overall (0.5, 0.2) 1
## 87      DR Overall (0.5, 0.3) 1
## 88      DR Overall (0.5, 0.4) 1

#####
# 7. Forest plot helper
#####

estimator_colors <- c(
  "IPW1" = "#009E73",
  "IPW2" = "#0072B2",
  "G-comp" = "#56B4E9",
  "DR" = "#E69F00"
)

estimator_shapes <- c(
  "IPW1" = 15,
  "IPW2" = 16,
  "G-comp" = 17,
  "DR" = 18
)

estimator_offsets <- c(
  "IPW1" = 0.27,
  "IPW2" = 0.09,
  "G-comp" = -0.09,
  "DR" = -0.27
)

make_effect_forest_plot <- function(plot_data, effect_name) {
  plot_data <- plot_data %>%
    filter(effect == effect_name)

  contrast_map <- plot_data %>%
    distinct(contrast, display_alpha1, display_alpha0) %>%
    arrange(desc(display_alpha1), desc(display_alpha0)) %>%
    mutate(contrast_id = row_number())

  plot_data <- plot_data %>%
    left_join(
      contrast_map,
      by = c("contrast", "display_alpha1", "display_alpha0")
    ) %>%
    mutate(
      y_pos = contrast_id + estimator_offsets[as.character(estimator)]
    )

  ggplot(
    plot_data,
    aes(color = estimator, shape = estimator)
  ) +
    geom_vline(

```

```

    xintercept = 0,
    linetype = "dashed",
    linewidth = 0.6,
    color = "gray30"
  ) +
  geom_segment(
    aes(x = lower, xend = upper, y = y_pos, yend = y_pos),
    linewidth = 0.8
  ) +
  geom_point(
    aes(x = estimation, y = y_pos),
    size = 2.2
  ) +
  scale_y_continuous(
    breaks = contrast_map$contrast_id,
    labels = contrast_map$contrast
  ) +
  scale_color_manual(values = estimator_colors, breaks = estimator_levels) +
  scale_shape_manual(values = estimator_shapes, breaks = estimator_levels) +
  labs(
    title = paste(effect_name, "Effects"),
    x = "Estimate (95% CI)",
    y = expression(paste("Allocation (", alpha[1], ", ", alpha[0], ")")),
    color = "Estimator",
    shape = "Estimator"
  ) +
  theme_minimal(base_size = 16) +
  theme(
    legend.position = "top",
    plot.title = element_text(face = "bold", hjust = 0),
    axis.text.y = element_text(size = 12),
    panel.grid.minor = element_blank(),
    panel.grid.major.y = element_blank()
  )
}

#####
# 8. Create four forest plots
#####

p_direct <- make_effect_forest_plot(df, "Direct")
p_indirect <- make_effect_forest_plot(df, "Indirect")
p_total <- make_effect_forest_plot(df, "Total")
p_overall <- make_effect_forest_plot(df, "Overall")

ggsave("spillover_forest_direct.pdf", p_direct, width = 10, height = 4.5)
ggsave("spillover_forest_indirect.pdf", p_indirect, width = 10, height = 5.5)
ggsave("spillover_forest_total.pdf", p_total, width = 10, height = 5.5)
ggsave("spillover_forest_overall.pdf", p_overall, width = 10, height = 5.5)

```